

Dashboard Design Template

Phase: Dashboard Design

Project Name: Food Consumer Behaviour Analysis

Team ID: PNT2022TMID01234

Date: 24 September 2024 / 15 October 2024

Maximum Marks: 5 Marks

Dashboard Design Principles & Overview

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience.

Activity 1: Interactive and Visually Appealing Dashboard Principles

- **Clear and Intuitive Layout:** Organized using a top-to-bottom F-pattern hierarchy with executive KPI summary cards on top, cross-sectional distributions in the center, and granular cohort trends at the base.
- **Use Appropriate Visualizations:** Selected tailored charts (bar charts for discrete category comparison, time-series line charts for seasonality, scatter plots for spend-satisfaction correlation, and donut charts for channel shares).
- **Colour and Theming:** Applied a polished, accessible color palette featuring Slate Navy, Electric Cyan, Emerald, and Coral ensuring high contrast and zero visual clutter.
- **Interactive Filters and Slicers:** Implemented synchronized global slicers for Age Cohort, Income Tier, Ordering Channel, Cuisine Family, and Date Range allowing effortless slice-and-dice.
- **Drill-Down Capabilities:** Enabled bidirectional Tableau action filters where selecting any cohort immediately cross-filters all corresponding visualizations.
- **Responsive Design:** Engineered dynamic canvas sizing supporting widescreen desktop displays (880px-1050px) down to mobile-friendly touch viewports.
- **Custom Visuals and Icons:** Integrated Lucide vector iconography and custom metric badge callouts for instantaneous visual comprehension.
- **Use of Infographics:** Incorporated visual summary banners, milestone progress bars, and percentage share callouts to enhance executive communication.

Major Outcomes from the Dashboard

- **Customer Base & Financial Scale:** The dashboard analyzes 10,000 active consumers representing \$2.30M in cumulative food expenditure, an average ticket size of \$38.40, and an aggregate customer satisfaction score of 4.35 / 5.0.
- **Dominant Ordering Channel:** Digital Food Delivery Apps lead overall transaction volume with 54.2% market share (\$1.25M sales), followed by Dine-in at 29.6% (\$680K) and Takeout at 16.2% (\$370K).
- **Top Revenue-Generating Cuisines:** Fast Food and Asian cuisines generate the highest total sales (\$620K and \$540K respectively), while Healthy & Organic dining records the highest average profit margin per meal (34.5%).
- **Peak Ordering Seasonality:** Monthly spending peaks strongly in November (\$320K) and December (\$352K) driven by holiday gatherings, year-end celebrations, and platform promotional events.
- **Demographic Spending Concentration:** The Young Professional & Millennial cohort (ages 25-34) constitutes 48.6% of all orders and exhibits the highest digital app adoption (72%).
- **Service Speed Impact on Loyalty:** Orders fulfilled within 30 minutes yield a 91% repeat order probability, whereas transit delays over 45 minutes trigger a 65% drop in Net Promoter Score.

Activity 2: Tableau Public Link

Dashboard Public URL:

<https://public.tableau.com/views/Foodconsumerbehaviouranalysis/FoodConsumerBehaviorAnalysis>